Today:

- classifying recurrence / transience for finite S.

- irreducibility & closed sets.

  - $A \subset S$ irreducible if $x \to y$ & $y \to x$ $\forall x, y \in A$

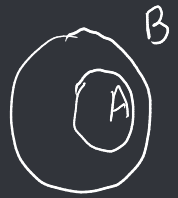
  - $A \subset S$ closed if $\forall x \in A$, $x \not\to y$ $\forall y \in A^c$.

- What's different when $|S| = \infty$?

- stationary distributions.

(1) Transience of x: $x \to y$ but $y \not\to x$:

Let $x = x_0, \dots, x_k = y$ such that



$$P(X_i = x_i \ \forall i \le k) > 0$$

& $x_i \neq x \ \forall i > 0$ so that

if $\underbrace{X_i = x_i \ \forall i \le k}_{A}$ then $\underbrace{T_x^+ = \infty}_{B}$

(if $T_x^+ > k$ and $X_k = y$ then since $y \not\to x$, $T_x^+ = \infty$).

Then $\rho_{x,x} \leq 1 - P(X_i = x_i \ \forall i \leq k) < 1$

$\therefore$ x transient.

② Recurrence w.r.t. if x: $\left.\begin{array}{c} x \to y \\ \text{implies} \\ y \to x \end{array}\right\} (*)$

& $|S| < \infty$ then x recurrent.

note $A_x := \{y : x \to y\}$ is closed.

In particular, $P(X_n \in A_x \mid X_0 = x) = 1$

$\forall n.$

If $y \in A_x$, by assumption $y \to x$

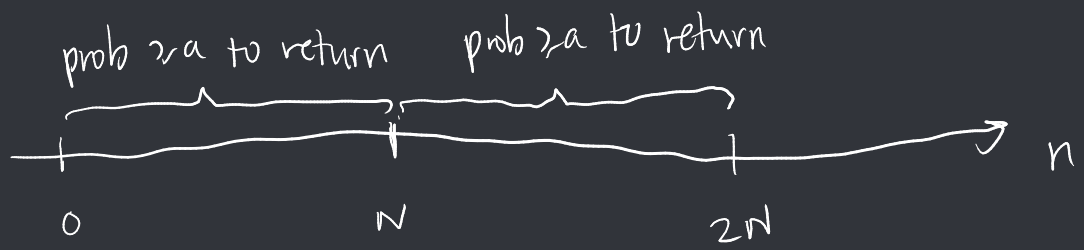
so $\exists\ n(x,y) :$ $p_{n(x,y)}(x,y) > 0$

$$\left.\begin{aligned} \text{Let } N &= \max_{y \in A_x} n(x,y) < \infty \\ a &= \min_{y \in A_x} p_{n(x,y)}(x,y) > 0 \end{aligned}\right\} \text{uses } |S| < \infty.$$

and $P(T_x^+ \le N \mid X_0 = y) \geq a \quad \forall y \in A_x$

Using Markov property, can show for $k = 1, 2, \ldots$

$$P(T_x^+ > kN \mid X_0 = x) \le (1-a)^k$$



intervals till first return $\le$ geom(a).

as $k \to \infty$,

$$P(T_x^+ = \infty) \leq \lim_{k \to \infty} P(T_x^+ > kN) = 0.$$

$\therefore$ x recurrent.

Classification of states.

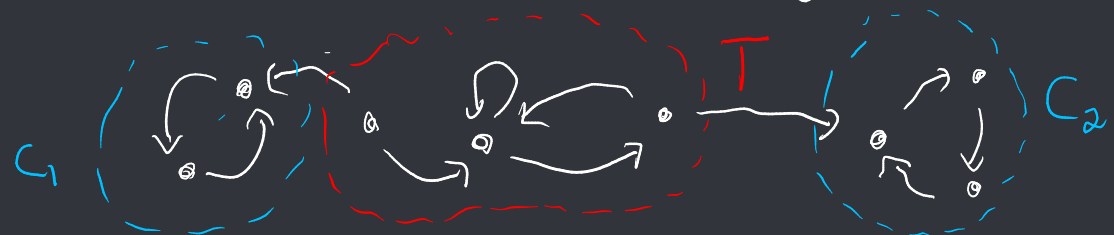
for a MC on S, $\exists$ a partition

$$\underbrace{C_1, \dots, C_k}_{\text{irreducible closed sets.}}, \underbrace{T}_{\text{transient states}}$$

$$T = \{ x : \exists\, y : x \to y \;,\; y \not\to x \}$$

each $C_i = A_x$ for any $x \in C_i$.

ex:



Infinite state space : closed & irreducible $\not\Rightarrow$ recurrent states.

ex: (ξ_n) i.i.d.

$$X_n = \sum_{i=1}^{n} \xi_i$$

if $E[\xi_n] \neq 0$ then

all states are transient,

strong law of large numbers:

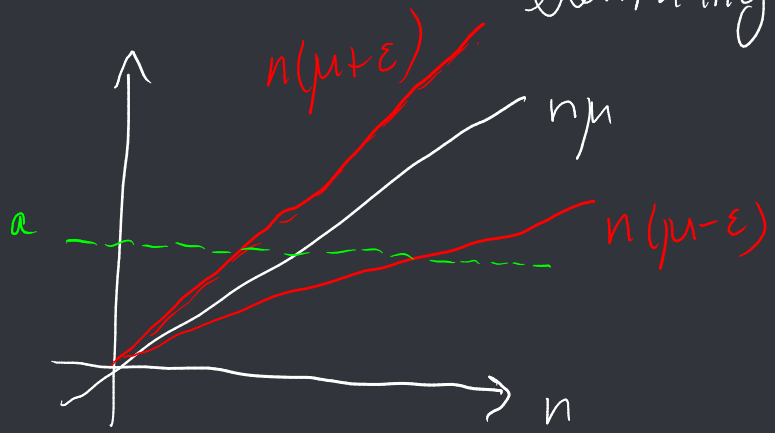
(1) $\frac{X_n}{n} - \mu \longrightarrow 0$ almost surely as $n \to \infty$.

$(\mu = E[\xi_n])$

(1) implies $\forall \varepsilon > 0$, eventually

$$|X_n - n\mu| \leq n\varepsilon.$$

if $\mu > 0$, say, then $\forall a$, eventually $X_n > a$.



Ex: "renewal" MC: $S = \mathbb{N}$ and

$$p(n,n+1) + p(n,0) = 1 \quad \forall n.$$

S

$\forall\, x, y \in S, \quad x \to y \;\&\; y \to x$

provided

$$p(n,n+1) > 0 \quad \forall n.$$

<u>Claim</u>: all states are transient if

$$\sum_{n=0} p(n,0) < \infty.$$

$$P(T_0^+ = \infty \mid X_0 = k) = \prod_{n=k}^{\infty} p(n, n+1)$$

$$= \prod_{n=k}^{\infty} (1 - p(n,0))$$

<u>Fact</u>: If $0 \le a_n < 1$ $\forall n$ then

$$\prod_{n=0}^{\infty} (1-a_n) = 0 \iff \sum_{n=0}^{\infty} a_n = \infty.$$

ex:

$$\tfrac{1}{2} \cdot \tfrac{2}{3} \cdot \tfrac{3}{4} \cdots = 0 \quad \& \quad \tfrac{1}{2} + \tfrac{1}{3} + \tfrac{1}{4} + \cdots = \infty.$$

Pf: bound $1-x$ by e^{-x}

$$\prod_{n=0}^{\infty}(1-a_n) \approx \prod_{n=0}^{\infty} e^{-a_n} = e^{-\sum_{n=0}^{\infty} a_n}$$

(quantitative bound to make it precise).

Returning,

$$P\left(T_0^+ = \infty \mid X_0 = k\right) > 0$$

$$\iff \sum_{n \geq k} p(n,0) < \infty$$

Stationary distribution

how long on average is spent in each state in the long run?

With $N_n(x) := \#\{m \le n : X_m = x\}$,

does

$$\lim_{n \to \infty} n^{-1} N_n(x)$$

exist & what does it equal?

almost surely

when it exists the values of $\pi(x) := \lim_{n \to \infty} n^{-1} N_n(x)$ are a stationary distribution (as long as X_0 is deterministic).

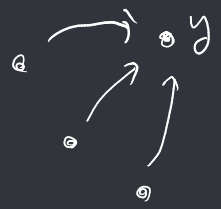
A distribution π on S is stationary if as a row vector

$$\pi = \pi p$$

transition matrix.

entrywise

$$\pi_y = \sum_x \pi_x \, p(x,y)$$



probab. @ y = incoming probab. to y in one step.

More generally for any initial distribution u_0

$$\left(P(X_0 = x) = u_0(x) \right),$$

the distribution

$$u_n(x) := P(X_n = x \mid X_0 \overset{d}{=} u_0)$$

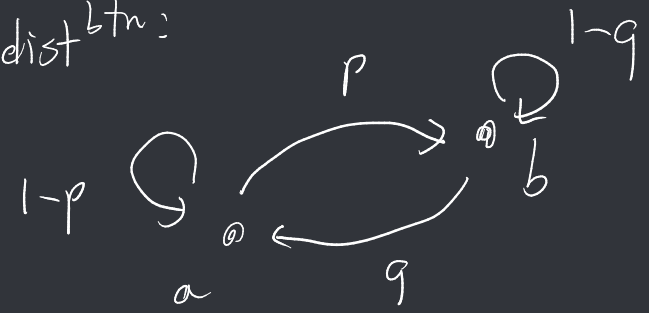
satisfies

$$u_n = u_0 P^n$$

$$\left(u_n(y) = \sum_x u_0(x) P(X_n = y \mid X_0 = x) \right)$$

so π is stationary $\Leftrightarrow \begin{cases} X_0 \overset{d}{=} \pi \\ \Rightarrow X_n \overset{d}{=} \pi \quad \forall n. \end{cases}$

<u>ex</u>: find stationary dist^{btn}:



want (π_a, π_b):

(1) $\pi_a = q\pi_b + (1-p)\pi_a$

(would do @ b as well; redundant)

also $\pi_a + \pi_b = 1$ (to be distbtn.)

(1) gives $p\pi_a = q\pi_b$ $\therefore$ $\pi_a = q/(p+q)$

$\pi_b = p/(p+q)$

<u>Claim</u>: if π stationary & $\pi_x > 0$ then x is recurrent.

<u>idea</u>: transient $\Rightarrow$ return only finitely many times.

$\pi_x > 0 \Rightarrow$ spend positive fraction time @ x in long run.

$E[\text{geo}(p)] = 1/p$.

(from geo dist. of # returns)

if x transient, $E_x[N(x)] = 1/(1-\rho_{xx})$

$$N(x) = \lim_{n\to\infty} N_n(x) = \#\{n : X_n = x\}$$

OTOH if π stationary & $\pi_x > 0$ then

$$E_\pi[N_n(x)] = (n+1)\pi_x \xrightarrow{n \to \infty} \infty$$

(uses $\underbrace{E_x[N_n(y)] = \sum_{m=0}^{n} p^m(x,y)}$)

$\forall$ events B, $P(B) = E[1(B)]$

proof: $p^m(x,y) = P_x(X_m = y)$

$= E_x[1(X_m = y)]$

where $1(A) = \begin{cases} 1 & A \text{ true} \\ 0 & A \text{ false.} \end{cases}$

and $\quad N_n(y) = \sum_{m=0}^{n} 1(X_m = y)$

then take $E[\cdots]$ & use linearity.

A state can be

transient $\Rightarrow$ 0 stat probab.

recurrent $\ldots = 0$ stat. probab.

> 0

prove later

$(= 0$ only if $|S| = \infty)$

Convergence of p^n, average frequency of visits.

If a MC is $\forall x, y \in S,\ x \to y,\ y \to x$

① → irreducible, aperiodic & has a stationary distribution π

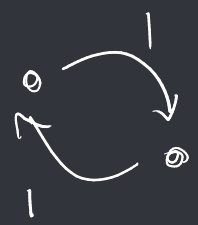
then $\forall x, y$ $\quad p^n(x,y) \xrightarrow{n \to \infty} \pi(y)$

② → irreducible & has a stationary distribution

then

$\forall$ distbtns. X_0, $\forall y$,

$$\frac{N_n(y)}{n} \longrightarrow \pi(y)$$ almost surely

as $n \to \infty$.

ex of 2, not 1: